Covariant Derivatives

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Abstract. This section introduces the definition of the derivative of a vector field, noting that its formulation relies inherently on the geometry of Euclidean space. The concept directly generalizes the directional derivative of a scalar function, $\mathbf{v}[f]$, at a point $\mathbf{p}$ along a tangent vector $\mathbf{v}$. By substituting the function with a vector field W, the field is evaluated along the straight line path $t \to \mathbf{p} + t\mathbf{v}$ to construct a vector field along a curve. Ultimately, the derivative of W with respect to $\mathbf{v}$ is defined by taking the standard calculus derivative of this restricted parameterization at $t = 0$.

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1. Covariant Derivatives

Definition 1.1 (Covariant Derivative)**.** Let W be a vector field on $\mathbf{E}^3$ and let $\mathbf{v}$ be a tangent vector to $\mathbf{E}^3$ at the point $\mathbf{p}$. Then the *covariant derivative of* W *with respect to* $\mathbf{v}$ is the tangent vector

$$\nabla_{\mathbf{v}} W = W(\mathbf{p} + t\mathbf{v})'(0)$$

at the point $\mathbf{p}$.

Notably, $\nabla_{\mathbf{v}} W$ measures the initial rate of change of $W(\mathbf{p})$ as $\mathbf{p}$ moves in the $\mathbf{v}$ direction.

Example 1.2. Compute the covariant derivative of the vector field $W = x^2U_1 + yzU_3$ with respect to the tangent vector $\mathbf{v} = (-1, 0, 2)$ at the point $\mathbf{p} = (2, 1, 0)$.

Solution. First, parameterize the straight-line path passing through $\mathbf{p}$ in the direction of $\mathbf{v}$ as $\mathbf{p} + t\mathbf{v} = (2 - t, 1, 2t)$. Next, evaluate the vector field W along this path by substituting $x = 2 - t$, $y = 1$, and $z = 2t$. Finally, compute the covariant derivative by evaluating the standard derivative at $t = 0$:

$$\begin{aligned} W(\mathbf{p} + t\mathbf{v}) &= (2-t)^2U_1 + (1)(2t)U_3 \\ &= (2-t)^2U_1 + 2tU_3 \end{aligned}$$

$$\begin{aligned} \nabla_{\mathbf{v}} W &= \frac{d}{dt}\Big|_{t=0} \Big[(2-t)^2U_1 + 2tU_3\Big] \\ &= -2(2-t)U_1 + 2U_3\Big|_{t=0} \\ &= -4U_1(\mathbf{p}) + 2U_3(\mathbf{p}) \end{aligned}$$

Lemma 1.3. *If $W = \sum w_i U_i$ is a vector field on $\mathbf{E}^3$, and $\mathbf{v}$ is a tangent vector at $\mathbf{p}$, then*

$$\nabla_{\mathbf{v}} W = \sum \mathbf{v}[w_i] U_i(\mathbf{p})$$

Theorem 1.4. *Let $\mathbf{v}$ and $\mathbf{w}$ be tangent vectors to $\mathbf{E}^3$ at $\mathbf{p}$, and let Y and Z be vector fields on $\mathbf{E}^3$. Then for all numbers a and b, and all (differentiable) functions f:*

$$\begin{aligned}
&(1) \quad \nabla_{a\mathbf{v}+b\mathbf{w}} Y = a\nabla_{\mathbf{v}} Y + b\nabla_{\mathbf{w}} Y \\
&(2) \quad \nabla_{\mathbf{v}}(aY + bZ) = a\nabla_{\mathbf{v}} Y + b\nabla_{\mathbf{v}} Z \\
&(3) \quad \nabla_{\mathbf{v}}(fY) = \mathbf{v}[f]Y(\mathbf{p}) + f(\mathbf{p})\nabla_{\mathbf{v}} Y \\
&(4) \quad \mathbf{v}[Y \cdot Z] = \nabla_{\mathbf{v}} Y \cdot Z(\mathbf{p}) + Y(\mathbf{p}) \cdot \nabla_{\mathbf{v}} Z
\end{aligned}$$

By applying the pointwise principle, we can extend the covariant derivative so that we differentiate a vector field W with respect to another vector field V instead of a single vector.

The result is a new vector field $\nabla_V W$ whose value at each point $\mathbf{p}$ is defined by $\nabla_{V(\mathbf{p})} W$. If $W = \sum w_i U_i$, it follows that:

$$\nabla_V W = \sum V[w_i] U_i$$

Example 1.5. Let $V = (y - x)U_1 + xyU_3$ and $W = x^2 U_1 + yzU_3$. Compute the general vector field $\nabla_V W$, and evaluate it at the point $\mathbf{p} = (2, 1, 0)$.

Solution. Using the coordinate identity $U_i[f] = \frac{\partial f}{\partial x_i}$, we apply the components of V to the components of W:

$$\begin{aligned}
V[x^2] &= (y - x)U_1[x^2] = 2x(y - x) \\
V[yz] &= xyU_3[yz] = xy^2 \\
\implies \nabla_V W &= 2x(y - x)U_1 + xy^2 U_3
\end{aligned}$$

To evaluate this field at $\mathbf{p} = (2, 1, 0)$, we substitute $x = 2, y = 1, z = 0$:

$$\begin{aligned}
(\nabla_V W)(\mathbf{p}) &= 2(2)(1 - 2)U_1(\mathbf{p}) + (2)(1)^2 U_3(\mathbf{p}) \\
&= -4U_1(\mathbf{p}) + 2U_3(\mathbf{p})
\end{aligned}$$

This perfectly matches our earlier calculation using the single tangent vector $\mathbf{v} = V(\mathbf{p}) = (-1, 0, 2)_{\mathbf{p}}$.

Corollary 1.6. *Let V, W, Y, and Z be vector fields on $\mathbf{E}^3$. Then for all numbers a and b, and all functions f and g:*

$$\begin{aligned}
&(1) \quad \nabla_V(aY + bZ) = a\nabla_V Y + b\nabla_V Z \\
&(2) \quad \nabla_{fV+gW} Y = f\nabla_V Y + g\nabla_W Y \\
&(3) \quad \nabla_V(fY) = V[f]Y + f\nabla_V Y \\
&(4) \quad V[Y \cdot Z] = \nabla_V Y \cdot Z + Y \cdot \nabla_V Z
\end{aligned}$$

References

[1] Barrett O'Neill. *Elementary Differential Geometry.* Academic Press, Revised 2nd Edition, 2006.